

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Dhanajeyan J () 192324211 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1.ANSWER

Introduction

An intelligent agent is a system that senses its environment and takes actions to achieve specific goals. In a smart home, it controls lighting, temperature, and security using sensors like motion detectors, temperature sensors, and cameras.

1. Simple Reflex Agent

- Works only on current sensor input using if-then rules.
- Example: If motion is detected, turn on the lights.
- Advantage: Fast and simple.
- Limitation: Cannot remember past events or learn user preferences.

2. Model-Based Agent

- Maintains an internal memory of previous states.
- Example: Remembers room occupancy before turning lights or AC on/off.
- Advantage: Makes better decisions using past information.
- Limitation: Does not optimize between different choices.

3. Goal-Based Agent

- Takes actions to achieve predefined goals.
- Example: Maintains a comfortable temperature while reducing energy usage.
- Advantage: Can plan actions to achieve objectives.
- Limitation: May not choose the most efficient option.

4. Utility-Based Agent

- Chooses the action that gives the highest overall benefit.
- Example: Balances comfort, energy efficiency, and security while learning user preferences.
- Advantage: Makes the best overall decision.
- Limitation: More complex to design.

Comparison

Agent Type	Memory	Planning	Learning
Simple Reflex	No	No	No
Model-Based	Yes	Limited	No
Goal-Based	Yes	Yes	Limited
Utility-Based	Yes	Yes	Yes

Best Approach

A Hybrid Agent (Model-Based + Goal-Based + Utility-Based) is the most effective for a smart home because it:

- Provides comfort by learning user preferences.
- Improves energy efficiency by reducing unnecessary power usage.
- Ensures security through quick responses and intelligent decision-making.

Conclusion

A hybrid intelligent agent is the best choice for a smart home automation system because it combines memory, planning, and optimization to provide better comfort, energy savings, and security.

2.ANSWER

Introduction

A robot placed in an unknown maze must find the exit without knowing the layout. Uninformed search algorithms help the robot explore possible paths until it reaches the goal.

1. Uniform Cost Search (UCS)

- UCS expands the node with the lowest path cost first.
- It guarantees the lowest-cost path to the exit.
- Example: If different paths have different movement costs, UCS chooses the cheapest route.

Advantages:

- Finds the optimal solution.
- Works well when path costs are different.

Disadvantages:

- High time complexity.
- Requires large memory to store explored nodes.

Complexity:

- Time: $O(b^{1+\lceil C^*/\epsilon \rceil})O(b^{1+\lfloor C^*/\epsilon \rfloor})O(b^{1+\lceil C^*/\epsilon \rceil})$
- Space: $O(b^{1+\lceil C^*/\epsilon \rceil})O(b^{1+\lfloor C^*/\epsilon \rfloor})O(b^{1+\lceil C^*/\epsilon \rceil})$

2. Iterative Deepening Search (IDS)

- IDS repeatedly performs Depth-First Search (DFS) with increasing depth limits until the goal is found.
- Suitable when the goal depth is unknown.

Advantages:

- Low memory usage.
- Complete for finite search spaces.
- Finds the shallowest solution.

Disadvantages:

- Repeats exploration of upper-level nodes.
- Not suitable when path costs differ.

Complexity:

- Time: $O(bd)O(b^d)O(bd)$
- Space: $O(bd)O(bd)O(bd)$

Intelligent Agents

Simple Reflex Agent

- Reacts only to current information.
- Not suitable because it cannot remember explored paths.

Model-Based Agent

- Maintains memory of visited locations.
- Avoids revisiting the same paths and makes better decisions.

Goal-Based Agent

- Searches for actions that lead to the exit.
- Uses search algorithms like UCS or IDS to reach the goal.

Best Combination

A Goal-Based Agent with Uniform Cost Search (UCS) is the most effective because:

- The goal-based agent focuses on reaching the exit.
- UCS guarantees the lowest-cost and optimal path.
- If memory is limited and all paths have equal cost, Goal-Based Agent with IDS is a good alternative.

Conclusion

UCS provides the optimal path but requires more memory, while IDS uses less memory and is suitable when the goal depth is unknown. A Goal-Based Agent with UCS is the best choice for finding the safest and most efficient path in an unknown maze.